Elements of AIML Lab
Experiment 10 (After-submission Reference): Visualization with Matplotlib & Seaborn

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1 How to use this reference

This reference is meant to be shared **after you submit** Experiment 10. It focuses on two skills:

- choosing the right plot for the right variable type,
- communicating a clear message (labels, titles, readable scales).

2 A simple plot-selection cheat sheet

- Numeric distribution → histogram/KDE, boxplot, violin plot
- Categorical counts → bar plot / count plot
- Numeric vs numeric → scatter plot (add transparency if many points)
- Numeric vs categorical → boxplot/violin by category
- Many numeric relationships → correlation heatmap / pairplot
- Time series → line plot (time on x-axis)

3 Worked example (good plotting hygiene)

3.1 Step 1: load and set a clean theme

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
df = pd.read_csv("processed.csv") # or your dataset
```

3.2 Step 2: create a 5-plot portfolio (template)

1) Histogram + KDE.

```python
sns.histplot(df["num_col"], kde=True)
plt.title("Distribution of num_col")
plt.xlabel("num_col")
plt.ylabel("Count")
plt.savefig("01_hist.png", dpi=150, bbox_inches="tight")
plt.show()
```

2) Boxplot (outliers/spread).

```python
sns.boxplot(x=df["num_col"])
plt.title("Boxplot of num_col")
plt.savefig("02_box.png", dpi=150, bbox_inches="tight")
plt.show()
```

3) Bar/count plot (categorical counts).

```python
sns.countplot(data=df, x="cat_col")
plt.title("Counts of cat_col")
plt.xticks(rotation=30, ha="right")
plt.savefig("03_count.png", dpi=150, bbox_inches="tight")
plt.show()
```

4) Scatter plot (relationship).

```python
sns.scatterplot(data=df, x="num_col1", y="num_col2", alpha=0.6)
plt.title("num_col1 vs num_col2")
plt.savefig("04_scatter.png", dpi=150, bbox_inches="tight")
plt.show()
```

5) Correlation heatmap (numeric).

```python
corr = df.select_dtypes("number").corr()
sns.heatmap(corr, cmap="coolwarm", center=0, annot=False)
plt.title("Correlation heatmap")
plt.savefig("05_heatmap.png", dpi=150, bbox_inches="tight")
plt.show()
```

4 How to write strong observations

For each plot, write 2–3 lines:

- what the plot shows (variables + plot type),
- the main pattern (skew, clusters, trend, outliers),
- a possible next action (missing/outlier handling, transformation, feature idea).

5 Common mistakes (and fixes)

- **Missing labels/titles:** add `plt.xlabel`, `plt.ylabel`, `plt.title`.
- **Overplotting:** use `alpha` or smaller markers; sample if extremely large.
- **Misleading scales:** avoid changing axes without explaining; prefer consistent scales for comparisons.
- **Plotting categories as continuous:** use countplots/bar plots for categories.
- **Not saving figures:** always `savefig` for submissions.

Sources: Provost and Fawcett, *Data Science for Business.*;McKinney, *Python for Data Analysis.*;WEKA Documentation (Waikato Environment for Knowledge Analysis), accessed 2026-02-28.;Matplotlib and Seaborn documentation, accessed 2026-02-28.